เลขที่นั่งสอบ

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Final Examination of Semester 1
Subject: 392133 Physics III
Date: 1 December 2016

Year: 2016
Section: 15-16
Time: 08.00-10.00

Name...ID..Class................

Instructions

1. Cheating in the exam is considered an extremely serious offence which will result in expulsion from the university.
2. No documents are allowed to be taken inside the examination room.
3. Express your answer in English. Other languages will be discarded.
4. No calculators are allowed.
5. Electronic communication devices are NOT allowed in the examination room.
6. The examination has 7 pages (including this page), 6 problems and a total score of 60 points.
7. Write all your answers in the examination sheet.

You are not allowed to leave the exam room during the first one hour and 30 minutes after the beginning of the exam.

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You are not allowed to use the restroom during the exam except in case of an emergency.

1.1 **[5 points]** A 5.0-ft woman wishes to see a full length image of herself in a plane mirror. What is the minimum length mirror required?

1.2 **[5 points]** A man stands with his nose 8 cm from a concave shaving mirror of radius 32 cm. What is the distance from the mirror to the image of his nose?

2.1 **[5 points]** The object-lens distance for a certain converging lens is 400 mm. The inverted-image is three times the size of the object. Describe how to make the image five times the size of the object.

2.2 **[5 points]** A plano-convex glass (n = 1.5) lens has a curved side whose radius is 50 cm. To obtain the image size that is to be the same as the object size, find the object distance from the lens.

3.1 **[5 points]** An object is 20 cm to the left of a lens of focal length +10 cm. A second lens, of focal length +12.5 cm, is 30 cm to the right of the first lens. Find the distance between the original object and the final image.

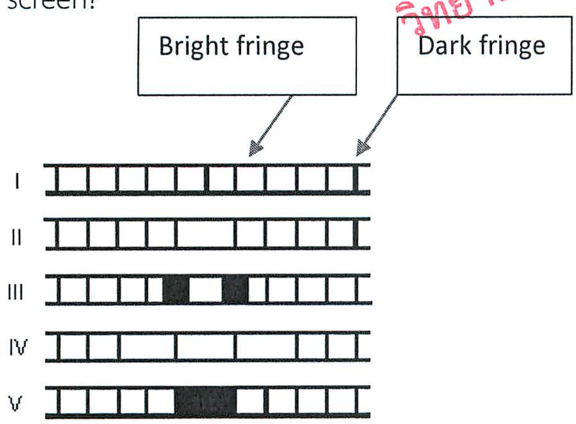
3.2 **[5 points]** A nearsighted person can see clearly only objects within 300 m of her eye. To see distant objects, she should wear eyeglasses of what type and focal length?

4.1 **[5 points]** A magnifying glass has a focal length of 15 cm. If the near point of the eye is about 25 cm from the eye, find the angular magnification (relaxing eye) of the glass.

4.2 **[5 points]** In a Young's double-slit experiment, light of wavelength 500 nm illuminates two slits which are separated by 1 mm. Find the separation between adjacent bright fringes on a screen 5 m from the slits.

5.1 **[5 points]** Monochromatic light, at normal incidence, strikes a thin film in air. If λ denotes the wavelength in the film, what is the thinnest film in which the reflected light will be a maximum?

5.2 **[5 points]** Monochromatic plane waves of light are incident normally on a single slit. Which one of the five figures below correctly shows the diffraction pattern observed on a distant screen?



6.1 **[5 points]** Monochromatic light is normally incident on a diffraction grating that is 1 cm wide and has 10,000 slits. The first order line is deviated at a 30° angle. What is the wavelength, in nm, of the incident light?

6.2 **[5 points]** Unpolarized light is passed through three successive Polaroid filters, each with its transmission axis at 45° to the preceding filter. What percentage of light gets through?

Asst. Prof. Dr. Amarin Ratanavis